Data Structures Assignment: Linear & Hierarchical Structures

This assignment covers three fundamental pillars of computer science: **Singly Linked Lists**, **Binary Search Trees (BST)**, and **Big O Notation**.

Question 1: Singly Linked Lists

Task: Define a Singly Linked List, explain its structure, provide a Python implementation for node insertion, and calculate the time complexity for searching an element.

Question 2: Binary Search Trees (BST)

Task: Explain the properties of a BST, provide the mathematical logic for its height, and show a recursive search function.

Question 3: Stack Operations & Postfix Evaluation

Task: Define the "LIFO" principle, explain the mathematical evaluation of a postfix expression using a stack, and provide the code for the push and pop operations.